A Large-Scale Multi-Modal Benchmark and Robust Text-Free Inference for Breast Ultrasound Video Segmentation

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1 Multi-modal dataset details

To promote research and development in the field of multi-modal breast tumor ultrasound video segmentation, this paper builds the first large-scale benchmark dataset for multi-modal breast tumor ultrasound video segmentation. The dataset contains 1085 breast ultrasound videos with a total of 50,375 finely annotated frames, covering BI-RADS [1] grades L2 to L5. Each case also has a paired diagnostic report.

All video data were collected by several professional doctors using the Mindray Resona 7 ultrasound machine. The data were annotated through several rounds of expert review, data cleaning, and quality testing steps. The annotation strategy is starting from the first frame of each video, every fifth frame is carefully labeled, to ensure high accuracy and consistency in tumor segmentation. All videos and diagnostic reports were anonymized to remove patient privacy information. The dataset is currently under ethical review and will be released after the paper is accepted.

This dataset is divided into training, validation, and test sets in a ratio of 6:2:2. The details of the split are shown in Table 1. To ensure the reproducibility of this research, the core code and dataset samples have been uploaded to an anonymous repository **(link: https://anonymous.4open.science/r/TF-BVS_code_and_dataset_demo-70D7)**. Specifically, the dataset visualization page is located in the **index.html** file within the repository, while the original data sample files are stored in the **assets** folder. Detailed explanations of the directory structure and instructions for viewing the dataset demo are provided in the **README.md** file. Some data examples are illustrated in Figure 2. All code and the complete dataset will be made publicly available upon the acceptance of this paper. Our dataset is **much larger** and **more diverse** than the open datasets available. **It provides a solid foundation and rich resources for future research in breast tumor ultrasound segmentation.** In the future, we will continue to expand and refine the dataset, with a particular focus on supplementing the scarce high-risk grade cases.

Table 1: We divided the dataset in a ratio of 6:2:2. All-Frame: The total number of frames in all videos. AnF: The number of finely annotated frames. Video: The number of videos. Text: The number of diagnostic reports paired with videos.

Set	All-Frame	AnF	Video	Text
Test	60769	10222	217	217
Valid	60425	10029	217	217
Train	181552	30124	651	651
All	302746	50375	1085	1085

2 Experimental Details

2.1 Data Preprocessing

Structured Diagnostic Reports. The original diagnostic reports were first manually de-identified before processing and then the locally deployed Lingshu-32B [2] model was used to parse the text into a structured format. We designed a prompt template for the model that includes fields such as exam site, nodule location, nodule size, shape, echogenicity, presence of calcification, and a description of surrounding tissue, as shown in Figure 1. If a report contains multiple nodules, the template instructs the model to list the details for each nodule in order. To ensure the reliability of the structured outputs, we manually checked the output for every sample.

Cross-frame Temporal Backbone. All input images are normalized using the specified mean and standard deviation. Random horizontal flipping is applied during training for data augmentation, improving model generalization.

Frame-Wise Temporal Branch. Dynamic sample segments are constructed based on each pair of adjacent annotated keyframes in every video. Each segment includes the start and end keyframes as well as all intermediate unannotated frames as temporal inputs. During training, random horizontal flipping is applied to all images for data augmentation. Additionally, during data loading, mask corruption is performed with a certain probability by erasing small regions of the segmentation mask. This simulates imperfect segmentation guidance and improves model robustness.

2.2 Setup

TF-BVS's cross-frame temporal branch strictly follows the configuration of ReferDINO and adopts a pretrained GroundingDINO as the backbone, where Swin Transformer is used for image feature extraction and BERT is used for text feature encoding. The model is trained on four RTX-4090 GPUs, 35 epochs on our multi-modal dataset and 150 epochs on the VTUS dataset. The AdamW optimizer is used, with a learning rate of 5×10^{-4} and a batch size of 2.

2.3 Comparative Experimental Setup for Natural-Scene Models

To ensure a fair comparison and effectively adapt existing natural-scene referring video object segmentation (RVOS) methods to the medical ultrasound domain, we standardized the input conditions for all baseline models.

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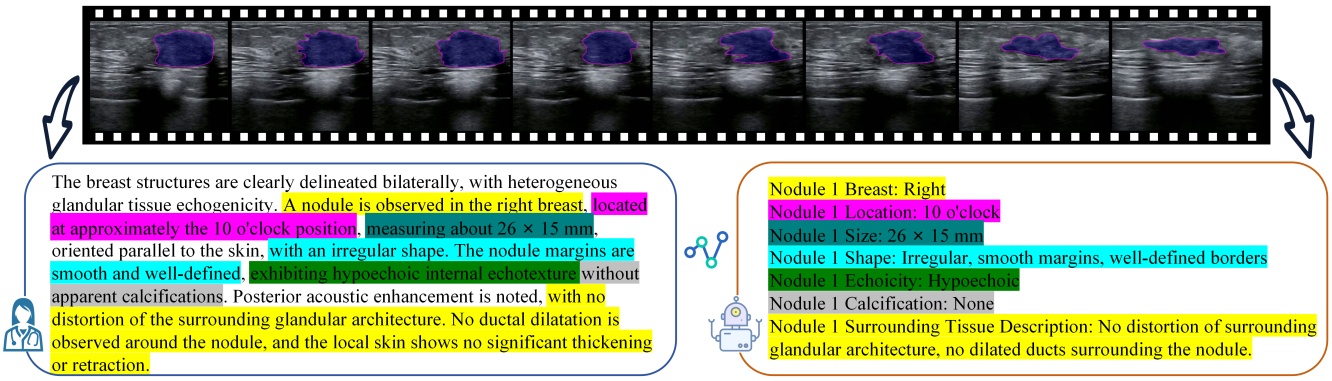


Figure 1: Example of using Lingshu32B [2] to perform text cleaning and structuring of original reports.

Specifically, for the text modality, rather than directly inputting raw, unstructured clinical diagnostic reports (which often contain substantial irrelevant domain noise), we utilized the locally deployed Lingshu-32B [2] model to pre-parse and clean the reports into a structured text format. This processing pipeline is consistent with the approach described in the main text. Subsequently, these cleaned, structured anatomical descriptions were directly used as the semantic text prompts for the baseline models. For the visual modality, all breast ultrasound videos were fed into the models following the standard data loading and normalization protocols inherent to each respective baseline method.

Importantly, we did not make any domain-specific architectural modifications to these natural-scene baselines, nor did we incorporate any additional medical prior modules. This unified and standardized processing pipeline ensures fairness, guaranteeing that the performance differences observed in our experiments strictly reflect the inherent capabilities and limitations of the models' original architectural designs when tackling complex multi-modal medical tasks.

2.4 Training

During training, the AdamW optimizer is used. The learning rate is scheduled with MultiStepLR and decays in stages according to predefined milestones, allowing joint scheduling of both backbone and branch modules. Automatic mixed precision (AMP) is applied throughout, combined with gradient scaling to improve memory efficiency and stability for large-batch training. Gradients are accumulated every N steps, and a maximum norm constraint is applied to prevent gradient explosion.

2.5 Inference

During inference, we first input a breast ultrasound video sequence. The model predicts the corresponding BI-RADS tumor grade label. Then, using the designed LV2TOT module, the video features are mapped into a set of pseudo-text token embeddings. These pseudo-text token embeddings are used as semantic prompts in the decoder to guide the segmentation. For the joint inference process, we adopt a two-stage inference strategy. First, the Cross-frame Temporal Backbone performs sparse sampling on the video with a predefined frame interval (for example, every k frames), and runs cross-frame inference only on these key frames. It produces the corresponding segmentation masks and query response scores. Based on the prediction scores on all sampled frames, the backbone selects the best query and outputs high-confidence segmentation results on the key frames. Next, the Frame-wise Temporal Branch performs segmentation on the entire video based on the backbone results. Specifically, for each sequence of frames defined by two adjacent key frames, we use the image of the starting key frame and its backbone-predicted mask as memory guidance, and perform roll-out inference frame by frame on all intermediate frames in the segment to generate continuous temporal segmentation results. At the same time, on each key frame, we apply gated fusion between the backbone mask and the branch prediction mask. A learnable fusion parameter is used to compute a weighted average of the two masks, which adaptively balances the robustness of the backbone and the temporal consistency of the branch. Finally, the sparse key frame predictions from the backbone and the dense intermediate frame predictions from the branch together form the final segmentation result of the entire video.

2.6 Evaluation Metrics

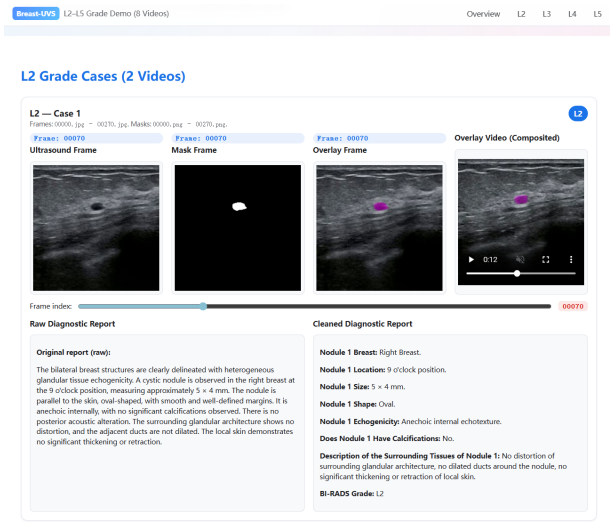
For quantitative evaluation of breast tumor ultrasound video segmentation, we compute several standard binary segmentation metrics on each video frame, including the Jaccard index (Intersection-over-Union, IoU), Dice coefficient, boundary F-measure, 95% Hausdorff distance (HD95), and Average Surface Distance (ASD). Let $G \subset \Omega$ and $P \subset \Omega$ denote the foreground regions of the ground-truth mask and the predicted mask, respectively, where Ω is the image domain. All masks are binarized before metric computation.

Jaccard Index (IoU). The Intersection-over-Union, also known as the Jaccard index, measures the overlap between the predicted and ground-truth regions:

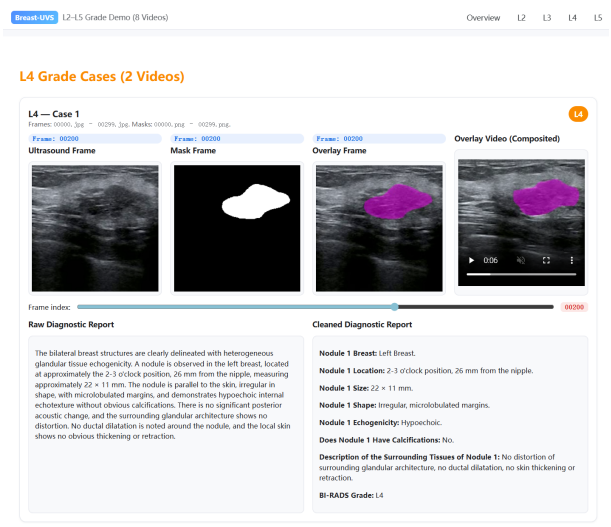
$$J(G,P) = \mathrm{IoU}(G,P) = \frac{|G \cap P|}{|G \cup P|}. \tag{1}$$

Dice Coefficient. The Dice coefficient evaluates the overlap between two sets:

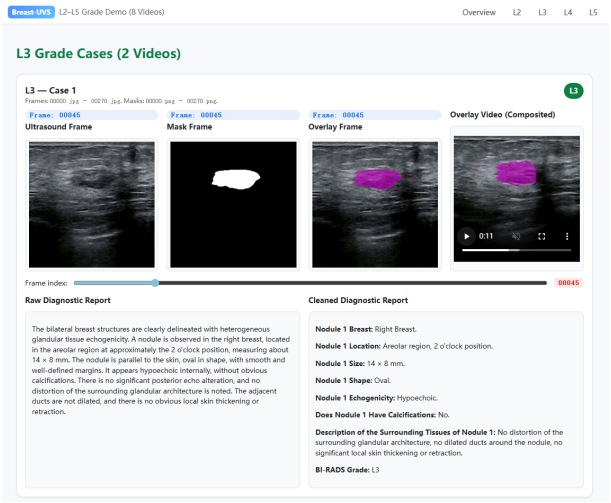
$$\mathrm{Dice}(G,P) = \begin{cases} 1, & \text{if } |G| + |P| = 0, \\ \dfrac{2\,|G \cap P|}{|G| + |P|}, & \text{otherwise.} \end{cases} \tag{2}$$



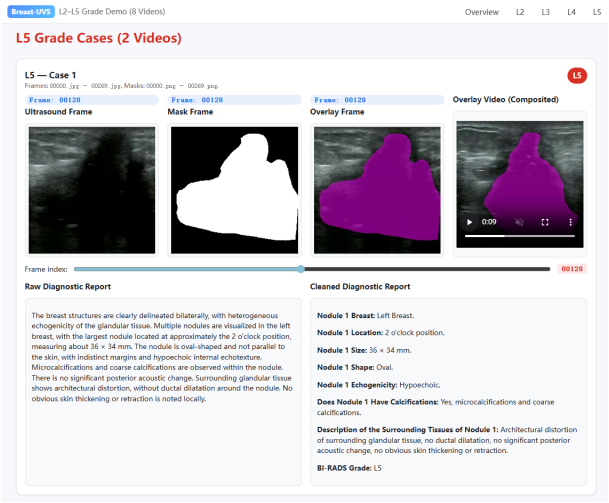
(a)



(b)



(c)



(d)

Figure 2: The figure shows the local demo page of our multi-modal breast tumor ultrasound video segmentation dataset. More interactive examples can be found in the index.html file and the assets folder in the anonymous repository.

Boundary F-measure. The boundary F-measure focuses on the alignment of object contours. We first extract the boundary maps B_G and B_P from G and P using a morphological boundary extraction operator. Given a boundary tolerance radius r (chosen as a fixed fraction of the image diagonal), we dilate each boundary map:

$$B_G^{(+r)} = \text{dilate}(B_G, r), B_P^{(+r)} = \text{dilate}(B_P, r). \tag{3}$$

The number of matched boundary pixels in each direction is

$$M_P = |B_P \cap B_G^{(+r)}|, M_G = |B_G \cap B_P^{(+r)}|. \tag{4}$$

Let $n_P = |B_P|$ and $n_G = |B_G|$. The boundary precision and recall are defined as

$$\text{Precision} = \begin{cases} 1, & n_P = 0 \text{ and } n_G > 0, \\ 0, & n_P > 0 \text{ and } n_G = 0, \\ 1, & n_P = 0 \text{ and } n_G = 0, \\ \dfrac{M_P}{n_P}, & \text{otherwise}, \end{cases} \tag{5}$$

$$\text{Recall} = \begin{cases} 0, & n_P = 0 \text{ and } n_G > 0, \\ 1, & n_P > 0 \text{ and } n_G = 0, \\ 1, & n_P = 0 \text{ and } n_G = 0, \\ \dfrac{M_G}{n_G}, & \text{otherwise.} \end{cases} \tag{6}$$

The boundary F-measure (F1-score) is then given by

$$F = \begin{cases} 0, & \text{Precision} + \text{Recall} = 0, \\ \dfrac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, & \text{otherwise.} \end{cases} \tag{7}$$

95% Hausdorff Distance (HD95). We compute the 95% Hausdorff distance between the predicted and ground-truth masks. Let S_G and S_P denote the sets of boundary points of G and P, respectively, and define the point-to-set distance

$$d(x, S) = \min_{y \in S} \|x - y\|_2. \tag{8}$$

The directed distance sets are

$$D(G \to P) = \{d(x, S_P) \mid x \in S_G\}, \tag{9}$$

$$D(P \to G) = \{d(y, S_G) \mid y \in S_P\}. \tag{10}$$

Let $Q_{0.95}(\cdot)$ denote the 95th percentile. The HD95 is defined as

$$\text{HD95} = \max\big(Q_{0.95}(D(G \to P)),\ Q_{0.95}(D(P \to G))\big). \tag{11}$$

Average Surface Distance (ASD). The Average Surface Distance measures the symmetric average distance between the two surfaces:

$$\text{ASD} = \frac{1}{|S_G| + |S_P|} \left(\sum_{x \in S_G} d(x, S_P) + \sum_{y \in S_P} d(y, S_G) \right), \tag{12}$$

where $d(\cdot, \cdot)$ is the Euclidean point-to-set distance defined above.

Frame-wise Aggregation. For each test video, we iterate over all frames whose ground-truth masks contain non-empty tumor regions, compute the above metrics on a per-frame basis, and then report the global mean of IoU (Jaccard index), boundary F-measure, Dice, HD95, and ASD across all evaluated frames.

3 Supplementary Experiment Result

Table 2 shows the parameter size and inference speed (FPS) of the TF-BVS model. The total number of parameters is about 22.59M, and the overall inference speed is 70.28 FPS. The model keeps the backbone accurate on key frame segmentation, while making the per-frame branch lightweight to improve the overall inference speed. In this way, it achieves a good balance between segmentation accuracy and real-time performance.

We compare several models with TF-BVS on our multi-modal dataset in terms of both inference speed and segmentation accuracy. In Figure 3, each point represents one method. The bubble area also reflects the Jaccard score. TF-BVS reaches the best Jaccard segmentation accuracy of 75.34% while achieving an inference speed of 70.28 FPS. This shows that the method can keep high segmentation quality and at the same time reach near real-time inference speed, and its overall performance is better than existing models.

Table 3 shows the ablation results of the key components of TF-BVS on our multi-modal dataset. The Baseline is a multi-modal video segmentation model that requires the paired diagnostic report during both training and inference. To verify the effect of text guidance we remove the text modality and keep only single-modal training and inference (UnText). The segmentation performance drops sharply, proving that the diagnostic text provides crucial semantic constraints for tumor localization and boundary discrimination and supplements missing information under sparse annotation. To address the textual reliance on inference, we first try a prototype-space method. We build a video–text prototype library from the training set and at inference, use the global video feature to retrieve an approximate text embedding to guide segmentation **(Q-A)**. This strategy partially relieves text dependence, but it loses segmentation accuracy and is sensitive to training set scale. In contrast, the Visual-To-Text Optimal Transport module **(V2TOT)** aligns the real text distribution using partially text-paired samples.

Table 2: Model parameters and inference speed.

Module	Parameters (M)	FPS
Cross-frame Backbone	16.26	67.02
Wise-frame Branch	6.33	89.07
Total	22.59	70.28

Table 3: Ablation and comparison experiments. TG: whether text-guided segmentation is used. TFI: whether no-text inference is supported.

Methods	TG	TFI	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	Dice↑
Baseline	✓	×	74.20	67.00	82.13
+UnText	×	×	63.63	53.80	72.40
+Q-R	✓	✓	72.30	65.90	80.74
+V2TOT	✓	✓	73.16	66.89	81.48
+LV2TOT	✓	✓	73.96	67.59	82.25
+AD	✓	×	74.31	67.12	82.17
+AD+LV2TOT	✓	✓	74.15	67.23	82.34
+AD+Class-Emb	×	×	67.37	60.25	76.07
+AD+Branch	✓	×	75.30	67.96	83.57
+AD+Branch+LV2TOT	✓	✓	**75.34**	**68.10**	**83.69**

Table 4: Number of video samples for each tumor grade in the multi-modal dataset.

Grade	Train	Valid	Test	All
L2	81	19	14	114
L3	413	142	146	701
L4	147	51	52	250
L5	10	5	5	20

Table 5: Classification accuracy (%) for each tumor grade on the test set.

L2	L3	L4	L5	Average
64.29	83.70	59.62	40.00	61.90

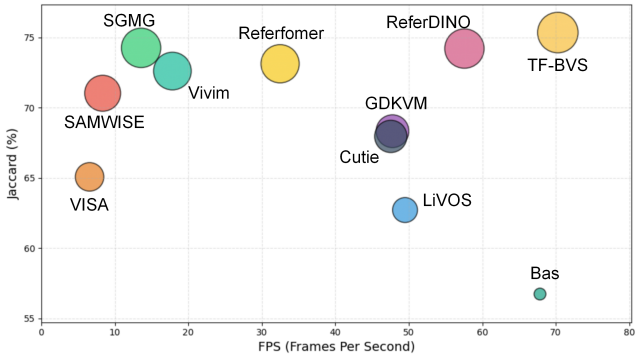


Figure 3: Compare the FPS and Jaccard scores of different methods on the multi-modal dataset.

It allows the model at inference to directly generate pseudo-text token embeddings from video features that follow the real diagnostic text distribution, thus reducing the performance gap to the baseline without requiring real text and easing dependence on data scale. Furthermore, the Label-aware Visual-To-Text Optimal Transport **(LV2TOT)** module injects tumor grade information into the cost matrix so videos of different grades map to their own pseudo text distributions. Compared to prototype retrieval, it produces more discriminative pseudo-text embeddings and clearly improves segmentation under text-free inference. On this basis, we design a dynamic adaptive gating mechanism in the memory update **(AD)** for the backbone, which dynamically adjusts memory retention and update to handle rapid tumor shape changes across key frames, improving temporal consistency and boundary detail. Furthermore, a simple learnable classification embedding (Class-Emb) baseline shows that merely injecting grade labels lacks rich lesion semantics, heavily bottlenecking performance by classification accuracy. This proves that our pseudo-text embeddings successfully filter report noise while delivering precise semantic guidance for segmentation. Finally, by combining the key-frame driven frame-wise temporal branch model **(Branch)** and the dual-branch cooperative training strategy, which form the complete TF-BVS framework, the cross-frame temporal backbone only segments key frames and initializes the branch memory while the lightweight branch propagates memory over intermediate unlabeled frames, achieving efficient segmentation and robust generalization under sparse annotation.

Table 4 shows the sample distribution of different tumor grades in the dataset. Among 1085 videos in total, L3 is the clear majority, while L2, L4, and L5 samples are much fewer, forming a typical long tail pattern. To reduce the effect of this imbalance on classifier training, we add class weighting to the loss in the classification, which improves learning for medium and low frequency classes. The grading classification accuracy of the model is shown in Table 5. The model achieves strong discrimination on the majority class L3, and keeps 64.29% and 59.62% accuracy on the two medium classes L2 and L4. It is important to note that this classification module mainly serves to provide a conditional prior for the LV2TOT module, rather than to pursue extreme accuracy for the grading task. During training LV2TOT uses ground truth grade labels to align class conditioned pseudo-text distributions, so it is not affected by

Table 6: Loss weight settings and search ranges.

Loss weight	Search ranges	Best value
λ_1	{3.0, 4.0, 5.0, 6.0, 7.0}	5.0
λ_2	{1.0, 2.0, 3.0, 4.0, 5.0}	3.0
λ_3	{1.0, 1.5, 2.0, 2.5, 3.0}	2.0
λ_4	{1.0, 1.5, 2.0, 2.5, 3.0}	2.0

Table 7: Adaptive gating mechanism hyperparameter settings and search ranges.

Hp	Search ranges	Best value
$\alpha_{\max}$	{0.20, 0.50, 0.60, 0.70}	0.60
$\alpha_{\min}$	{0.01, 0.05, 0.10, 0.15}	0.05
β_σ	{1.00, 2.00, 5.00, 10.00}	5.00
σ_0	{0.02, 0.05, 0.08, 0.10}	0.05
λ	{0.20, 0.25, 0.30, 0.35}	0.25

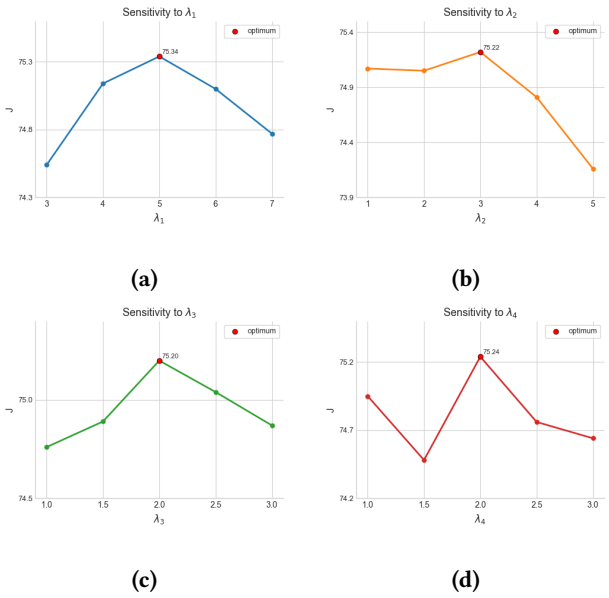


Figure 4: Sensitivity analysis of loss weights. (a) λ_1, (b) λ_2, (c) λ_3, (d) λ_4.

inference time classification errors, and even if the overall classification accuracy at inference is 61.90%, it still provides enough semantic bias to separate different grades, which effectively alleviates the text dependence problem. In future work can replace the classifier with a stronger or semi-supervised enhanced model to further improve the quality of pseudo-text embeddings for each grade. In addition, we will continue expanding and updating our multi-modal dataset by adding more cases for underrepresented tumor grades to provide a more balanced training basis and more reliable evaluation.

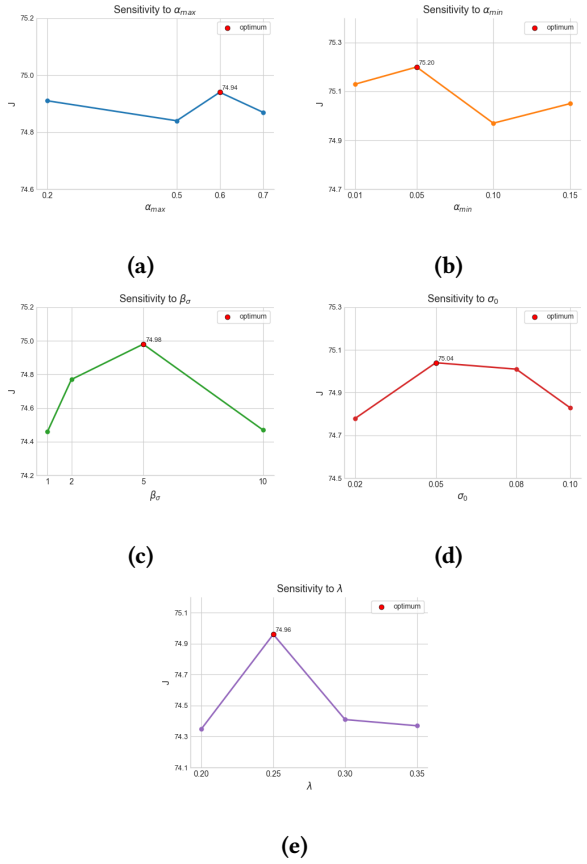


Figure 5: Sensitivity analysis of the adaptive gating mechanism. (a) $\alpha_{\max}$, (b) $\alpha_{\min}$, (c) β_σ, (d) σ_0, (e) λ.

4 Hyperparameter Sensitivity Analysis

In the experiment, we conducted a hyperparameter sensitivity analysis for the loss weights to evaluate their specific impact on the model's performance. Each weight was validated across a predefined set of five candidate values. To ensure the reliability and accuracy of the experimental process, we adjusted one weight at a time while keeping the other weights fixed. The search ranges for these weights and the final selected optimal values are detailed in Table 6, while the impact of these weights on performance is illustrated in Figure 4. The results further demonstrate that jointly optimizing the four loss terms can significantly enhance the cross-modal alignment between pseudo-text and real text.

The adaptive gating mechanism centers on the computation of the dynamic update rate α_{eff}, which is controlled by five critical hyperparameters: the maximum update rate $\alpha_{\max}$, the minimum update rate $\alpha_{\min}$, the sensitivity factor β_σ, the variance baseline σ_0, and the historical mean weight λ. These hyperparameters work in tandem to enable the model to dynamically adjust memory update rates based on inter-frame similarity variations, thereby achieving cross-frame object alignment and temporal consistency. To evaluate the influence of these hyperparameters on model performance, we conducted a sensitivity analysis. Each hyperparameter was validated across four candidate values while keeping the others fixed, ensuring the accuracy and reliability of the experimental results. The search ranges and optimal values are detailed in Table 7, while the corresponding impact on validation performance is illustrated in Figure 5. Results demonstrate that the adaptive gating mechanism plays a critical role in temporal prediction across frames. It enables the backbone model to respond swiftly to dynamic changes in tumor features, stabilizing the propagation of critical frame memory and providing reliable memory guidance for frame-by-frame predictions in the branch network.

References

[1] American College of Radiology. 2013. *ACR BI-RADS Atlas: Breast Imaging Reporting and Data System* (5th ed.). American College of Radiology, Reston, VA.

[2] Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, Ruifeng Yuan, Jianyu Wang, Chenghao Xiao, Guizhen Chen, Chaoqun Liu, Zhaodonghui Li, et al. 2025. Lingshu: A Generalist Foundation Model for Unified Multimodal Medical Understanding and Reasoning. *arXiv:2506.07044* (2025).